

Windows Terminal Navigation: Lab Manual

Beginner Course Handout

July 9, 2026

Lab Objectives

By the end of this lab, you will be able to:

1. Identify your exact location in the file system using the terminal.
2. View contents of any directory seamlessly.
3. Move forward, backward, and jump to the root directory using `cd`.
4. Create, copy, move, rename, and safely delete files and folders.
5. Leverage **Tab-Completion** to prevent typos.

Core Ground Rules (Read Before Proceeding!)

- 1. The Tab Key is Your Best Friend:** Never type out long folder names. Type the first 3 letters and press `TAB` to let the terminal autocomplete the name for you.
- 2. No Recycle Bin:** Files deleted using command lines skip the Windows Recycle Bin completely. They are **permanently destroyed**. Double-check before you hit Enter!

1 Task 1: Locating Yourself and Exploring

Before moving anywhere, you must know where you stand. Think of the terminal as a map where your current path is constantly listed.

1.1 Part A: Using Command Prompt (CMD)

1. Press the **Windows Key**, type `cmd`, and press Enter.
2. Look at the text to the left of your blinking cursor (e.g., `C:\Users\YourName>`). This is your **Current Working Directory**.
3. Type the following command to view all files and folders inside this directory:

```
1 dir
```

4. Press Enter. Note how folders are marked with `<DIR>`.

1.2 Part B: Using PowerShell

1. Press the **Windows Key**, type `powershell`, and press Enter.
2. Type the formal command to check your location:

```
1 Get-Location
```

3. Now type either of these short aliases to list the files and folders:

```
1 ls
```

2 Task 2: The Art of Movement with `cd`

`cd` stands for **Change Directory**. We will use it to travel through the folder structure hierarchy.

2.1 Step-by-Step Exercise

1. **Move Forward:** Look at the list generated from your last directory list command. Find the Desktop folder. Type the following and press Enter:

```
1 cd Desktop
```

2. **Step Backward:** You want to go back out to your main user folder. Type:

```
1 cd ..
```

3. **Go Deep in One Shot:** You can jump through multiple folders at once by providing the relative path:

```
1 cd Desktop\Documents
```

4. **Escape to the Root:** To drop everything and jump straight back to the base of your hard drive (C:\), type:

```
1 cd \
```

3 Task 3: Creating and Writing Sandbox Content

Let's build an isolated workspace on your Desktop to safely practice file mutations.

1. First, navigate back onto your Desktop (replace with your actual PC username):

```
1 cd C:\Users\YOUR_USERNAME\Desktop
```

2. **Create a Folder:** Create a brand new project directory named `TerminalSandbox`:

```
1 mkdir TerminalSandbox
```

3. **Step Inside:** Move into your newly created folder:

```
1 cd TerminalSandbox
```

4. **Create a File:** Create an empty text file named `notes.txt`:
CMD Command:

```
1 type nul > notes.txt
```

PowerShell Command:

```
1 New-Item notes.txt
```

4 Task 4: Duplicating, Moving, and Renaming

Now we will manipulate the items we just created inside our sandbox directory.

1. **Create a Subfolder:** Inside your current location, create a backup folder:

```
1 mkdir Archive
```

2. **Copy a File:** Duplicate your `notes.txt` file into the `Archive` folder:
CMD Syntax:

```
1 copy notes.txt Archive\
```

PowerShell Syntax:

```
1 Copy-Item notes.txt Archive\
```

3. **Move/Rename a File:** Let's take the original `notes.txt` file outside the `Archive` folder and rename it to `todo.txt`:

CMD Syntax:

```
1 move notes.txt todo.txt
```

PowerShell Syntax:

```
1 Move-Item notes.txt todo.txt
```

5 Task 5: The Cleanup Operation

Let's clean up our workspace. Remember, these commands bypass the Recycle Bin.

1. **Delete a File:** Erase the `todo.txt` file from your sandbox:

CMD: `del todo.txt`

PowerShell: `rm todo.txt`

2. **Force Delete Folder + Contents:** To completely purge the `Archive` folder along with everything inside it:

CMD:

```
1 rmdir /s /q Archive
```

PowerShell:

```
1 Remove-Item -Recurse -Force Archive
```

3. Step out of your sandbox folder back onto your main Desktop:

```
1 cd ..
```

4. Remove the empty sandbox directory completely to complete your lab:

```
1 rmdir TerminalSandbox
```

Final Check

Type `cls` (CMD) or `Clear-Host` (PowerShell) to wipe your terminal history screen completely clean before showing your instructor!